Geospatial Tools for Energy Access

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February 2025

“The views expressed in this material do not necessarily reflect the UK government’s official policies.”

UKaid from the British people

University Partnership:

University of Cambridge · The Open University · Climate Parliament · University of Strathclyde Glasgow · University of Oxford · Centre for Global Equality · UCL · Loughborough University (STEER – Centre for Sustainable Transitions: Energy, Environment and Resilience) · KTH Vetenskap och Konst · Imperial

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- OnStove Developments
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OnStove

OnSSET

Open Source Spatial Electrification Tool

OnStove – OnSSET Integration

A geospatial electrification and clean cooking model soft-link

An integrated planning approach for energy access

- Progress of SDG 7 targets has been uneven, 2021 in Kenya:
  - 77% of the population had access to electricity.
  - 28% of the population had access to clean cooking.
- An integrated approach of electricity and clean cooking can unlock the potential of electricity as clean cooking fuel.
- Without integrated planning:
  - 77% of the population would adopt electricity.
- With integrated planning:
  - Between 85 and 91% adopts electricity.

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Integrated geospatial modelling for the achievement of universal energy access in Kenya

Babak Khavari, Andreas Sahlberg, Camilo Ramirez, Sarah Odera, Elsie Onsongo, Kevin Nayema, Victor Otieno, Douglas Ronoh, Anobha Gurung & Francesco Fuso Nerini

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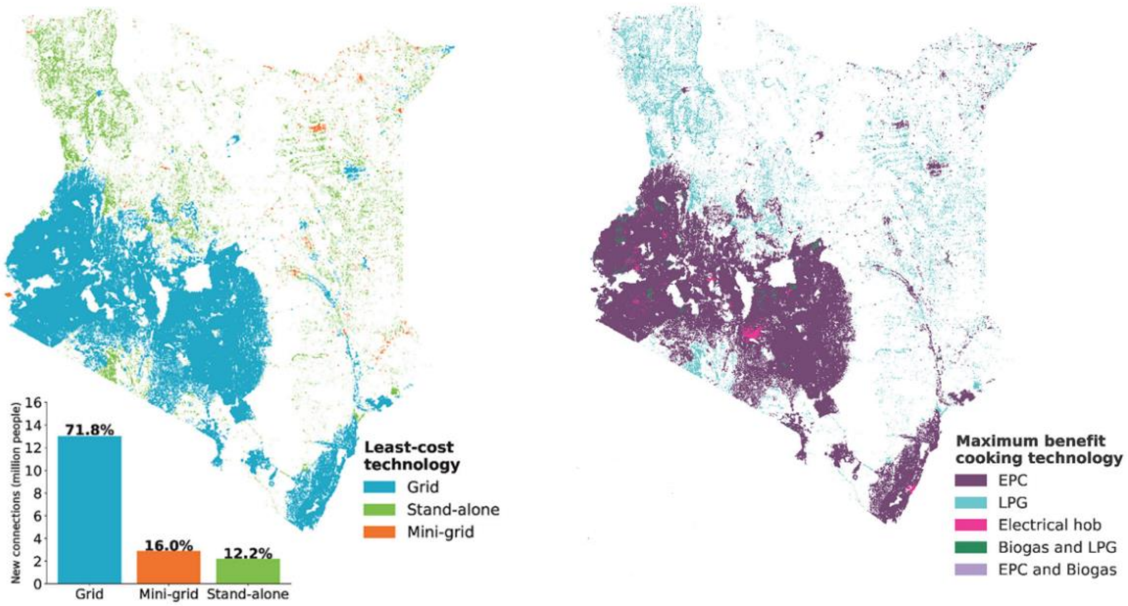
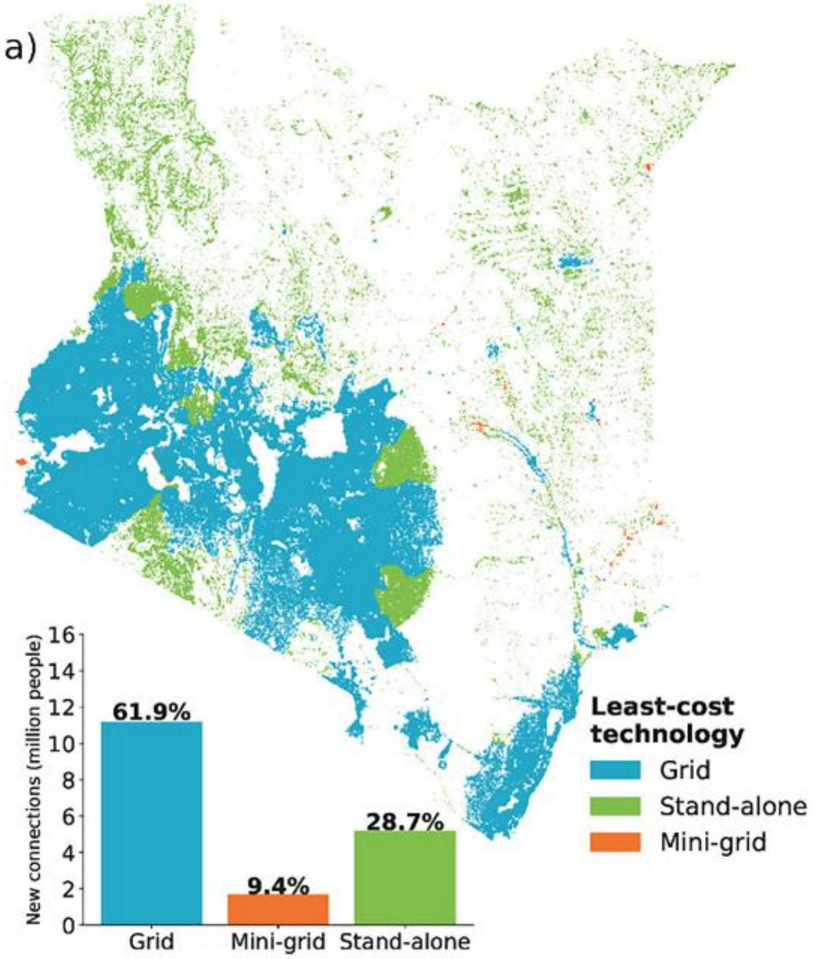
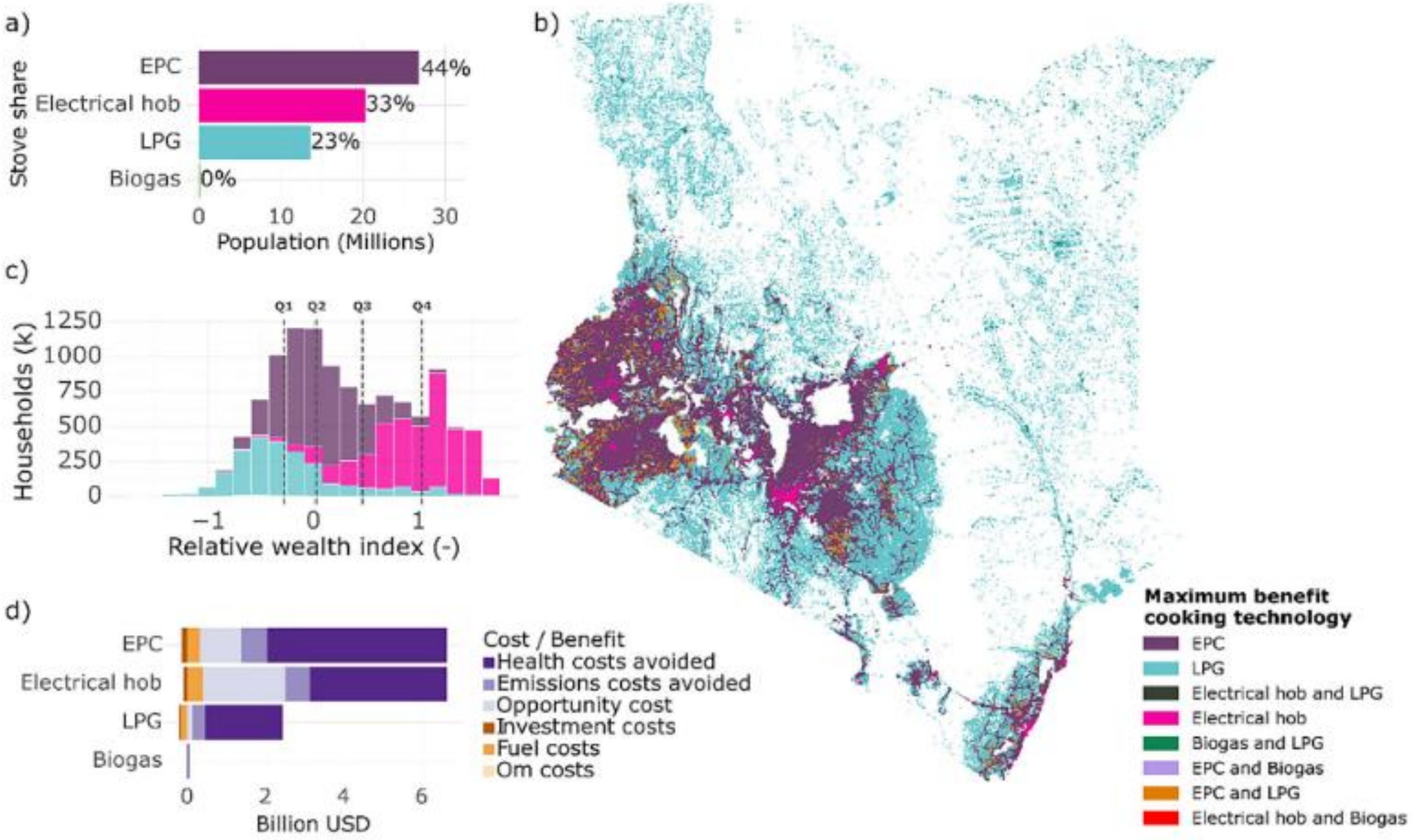


Energy Access without integrated planning

- In the study, electrification is achieved in 2026 and clean cooking in 2028 (start year 2021).



Electrification by OnSSET



Clean Cooking by OnStove

Energy Access with integrated planning

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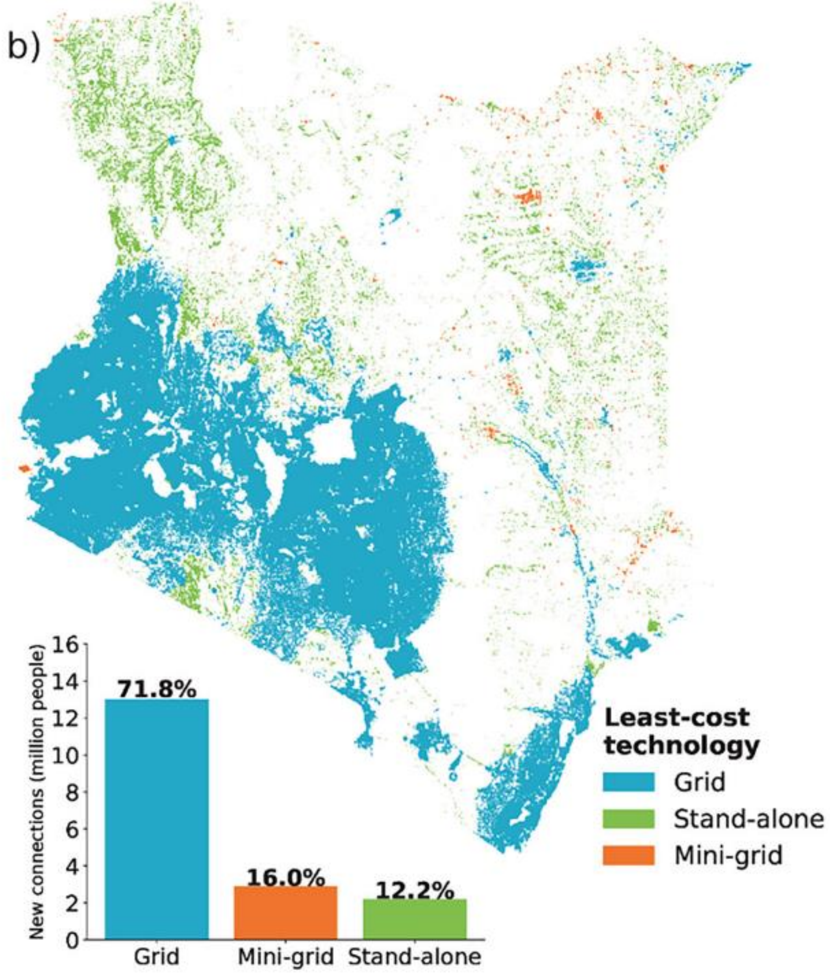
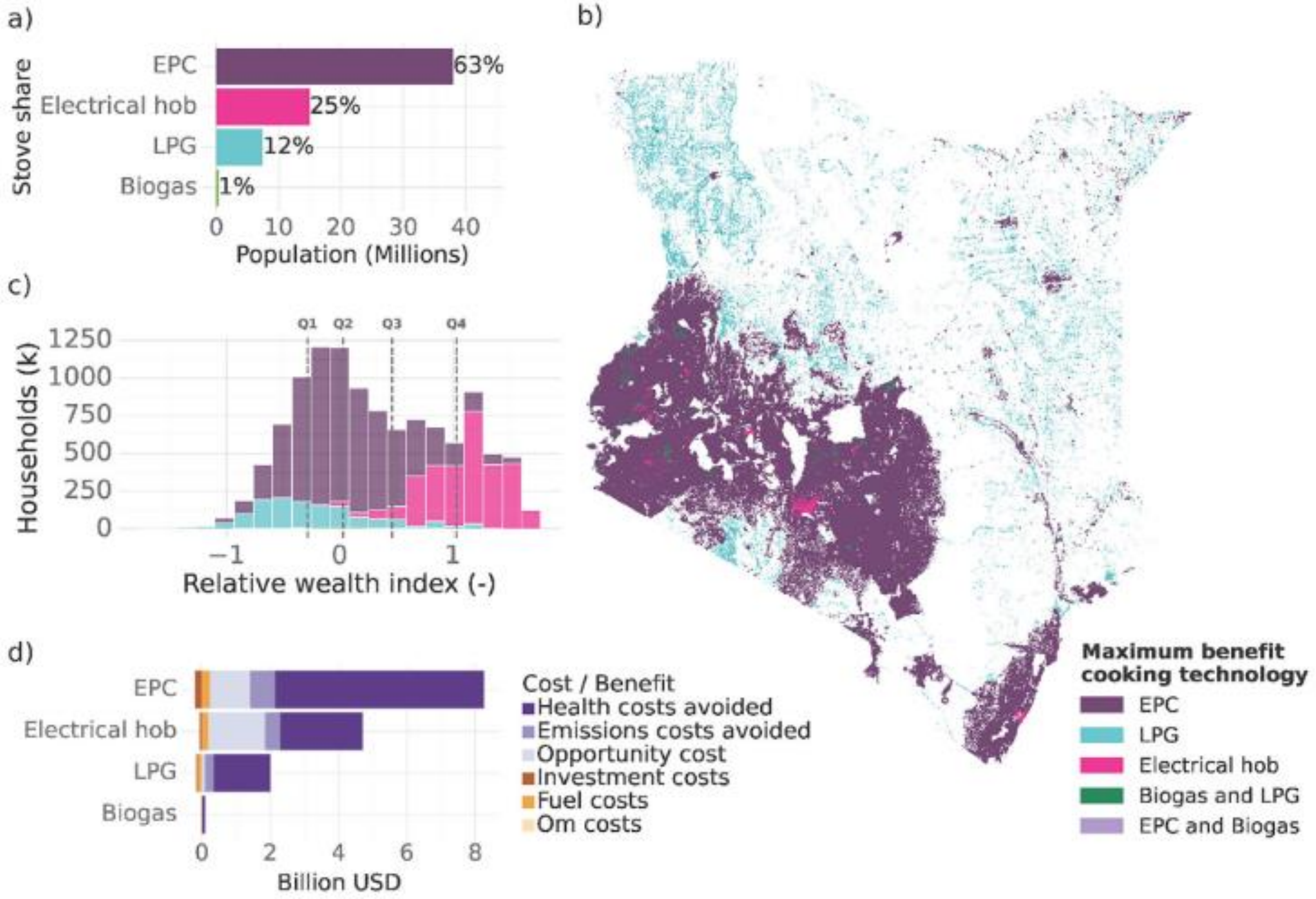


Electrification by OnSSET



Clean Cooking by OnStove

OnStove Developments

New features

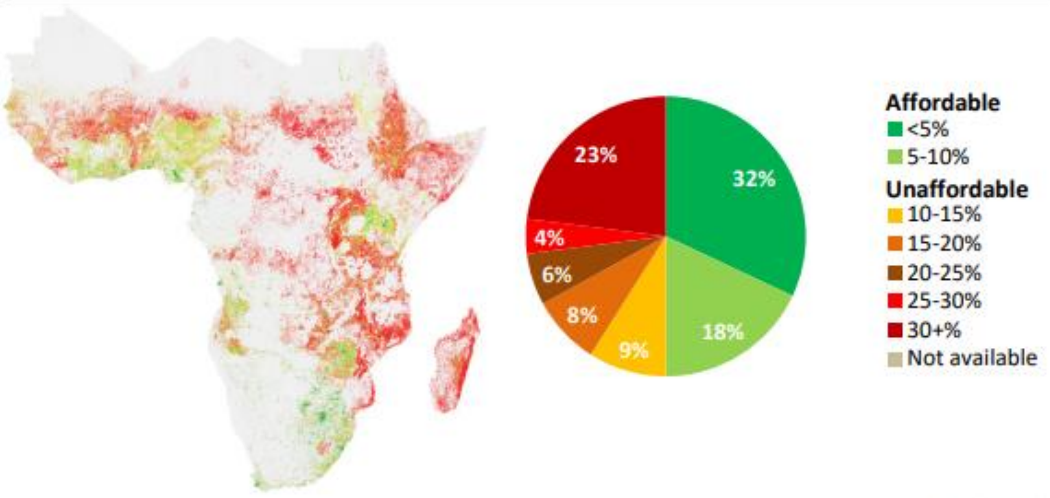
Is the solution presented by an OnStove scenario affordable?

- Incorporation of income estimations in the framework of OnStove.
- Leverages the Relative Wealth Index (RWI).
- Requires only GDP per capita and Gini coefficient as inputs.
- Can also work with other types of income distribution estimations.
- Other inputs are being considered, such as overlaying published GDP per capita dataset, for areas where RWI is not available.
- Puts more strain on the accuracy of the assumptions on energy per meal, household configuration, costs assumptions, and what is considerable affordable for cooking services.

Cost/income ratio – an affordability proxy

- **IEA's Universal Access to Clean Cooking in Africa Report**
  - Collaboration where both user-defined shares and cost/income maps were used as major points of analysis.

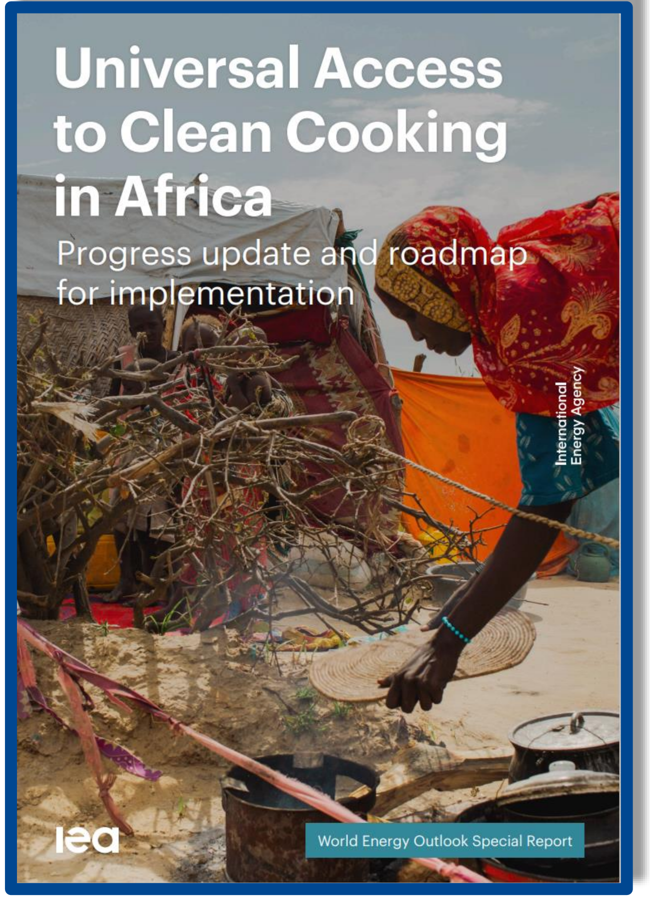
Figure 4.3 ▷ Affordability of LPG in sub-Saharan Africa, 2023



Coastal and river-corridor hubs keep LPG prices low by shaving up to 25% off logistics costs

Notes: The map displays where LPG is affordable, while the accompanying pie charts translate that same information into population shares. Map colours match the legend, with one exception: areas where electricity or biogas are unavailable are shown in grey on the maps. LPG = liquefied petroleum gas.

Sources: IEA and KTH Royal Institute of Technology analysis based on the OnStove model developed by KTH Royal Institute of Technology. Income levels per location are approximated based on the Relative Wealth Index (Meta, n.d.).

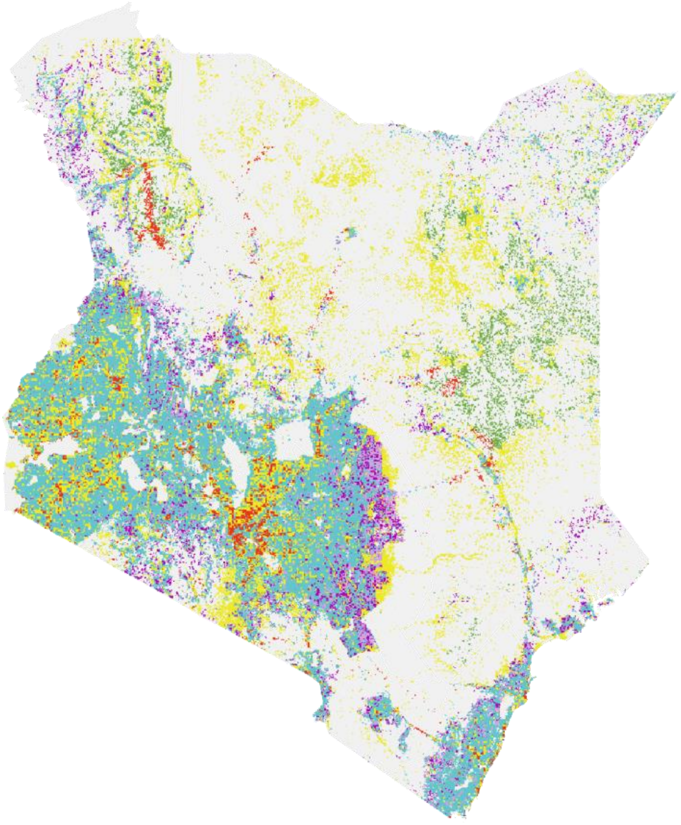


[Universal Access to Clean Cooking in Africa – Analysis - IEA]

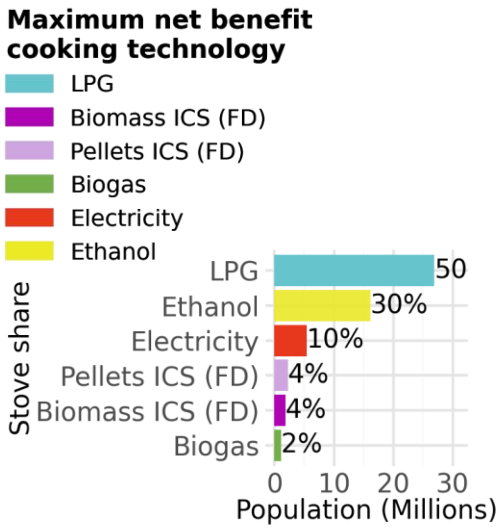
Can OnStove allocate stoves to achieve certain desired shares?

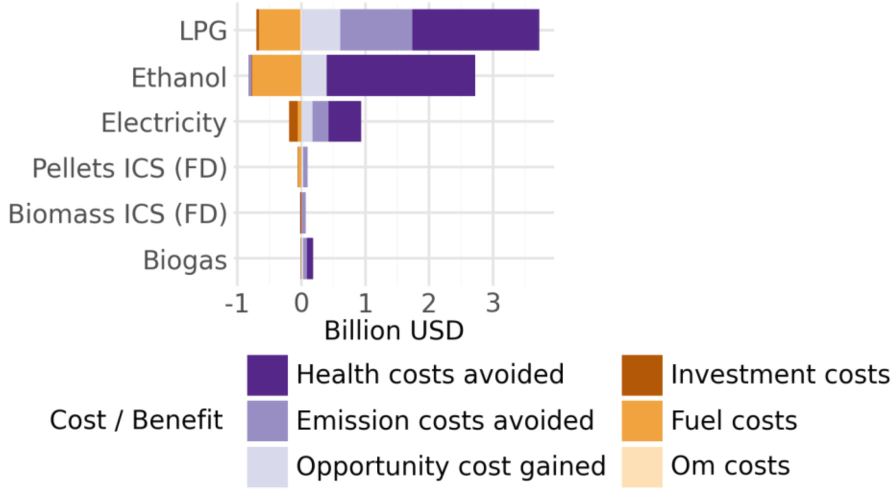
- **Preview of Results (Kenya's National Cooking Transition Strategy):**

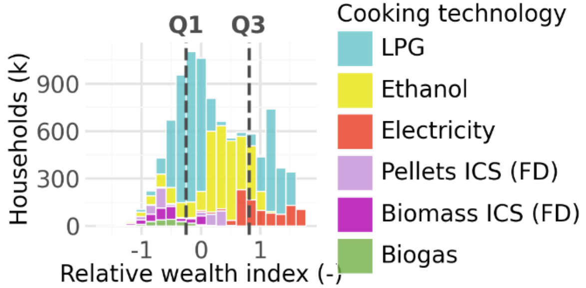
Maximum net benefit cooking technology



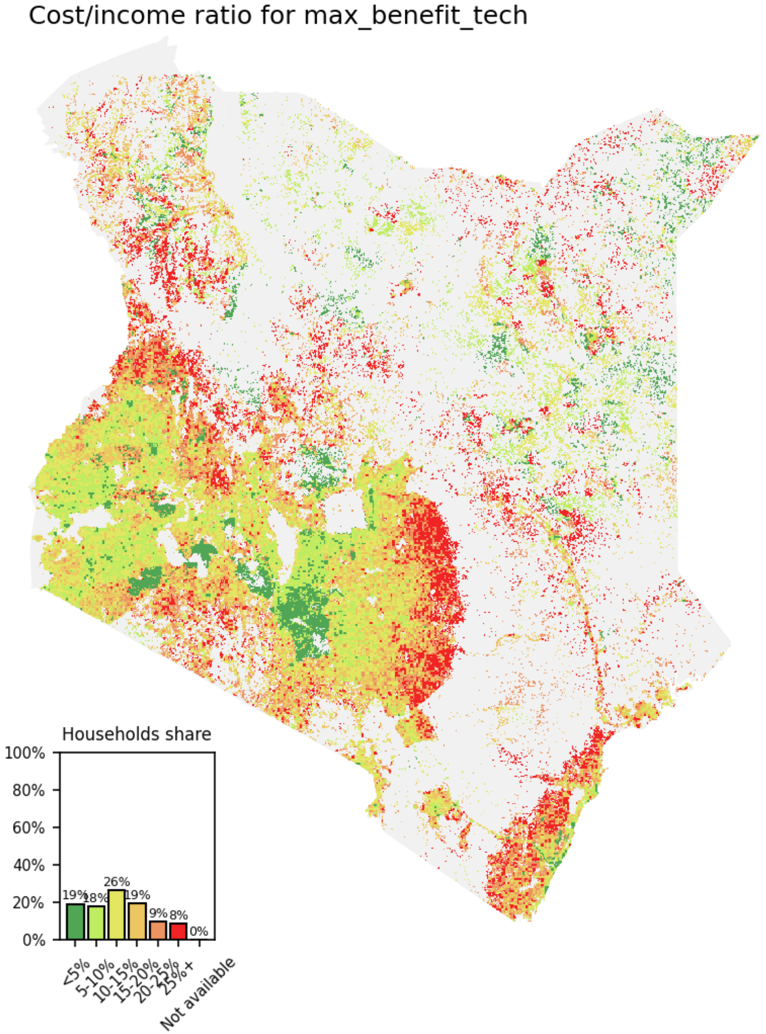
Deaths avoided	22,432 pp/yr
Health costs avoided	4.90 BUS$
Emissions avoided	29.40 Mton
Time saved	1.21 h/hh.day



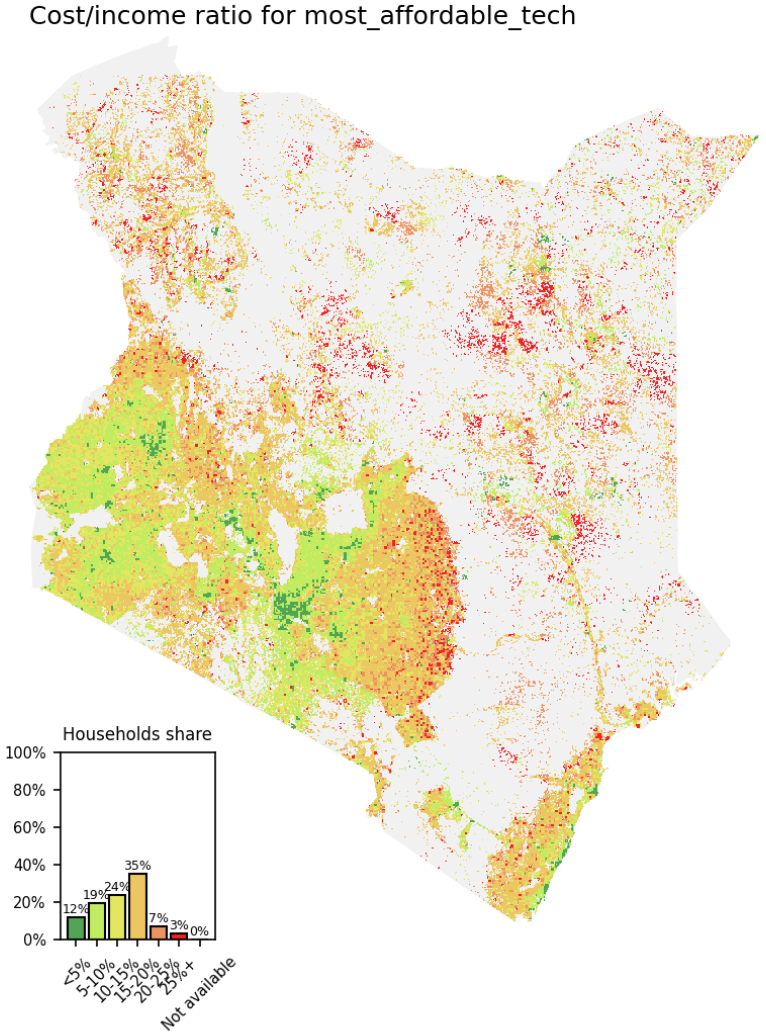




Together, both features enable new analytical approaches



Least-cost KNCTS



Affordable KNCTS

Current Work and Future Directions

Ongoing collaborations

- Climate Compatible Growth:
  - Development of starter kits for OnSSET and OnStove.
  - Research and software development
  - Participation in the Energy Modelling Platform
- UK Pact
  - Support for the development county-level energy plans, using OnSSET and OSeMOSYS
  - Development of a new methodology for accounting of fuel stacking in clean cooking
- International Energy Agency:
  - Supported the analyses of two reports.
  - Collaboration with the Ministry of Energy and Petroleum to develop a Kenya – specific OnStove model.

A Kenya specific OnStove model

- Cost components specific for the Kenyan context.
- Leveraging census data to the subcounty level (Administrative Level 2) (KNBS, 2019)
  - Baseline fuel use
  - Household average size
- Inclusion of mini-grid and Solar Home Systems locations in the calibration of electricity access.
  - This updates the places with electricity as an available technology.
- Feasibility / fact-checking of results
  - Against available county energy plans.
  - Against relationship of LPG consumption and LPG storage per province.

Current and future research

- Incorporation of institutional cooking
- Assessment of affordability
- Study of the LPG supply chain
- Discretization of the cooking demand according to cooking profile and fuel stacking
- Current baseline fuel-stove estimation and identification of solar home systems and mini-grids through machine learning
- Improvement of mini-grid representation in OnSSET

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Climate Compatible Growth

UK aid from the British people

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